Tutorial 3: Quasielastic Neutron Scattering (QENS)

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MDANSE Workshop 2026

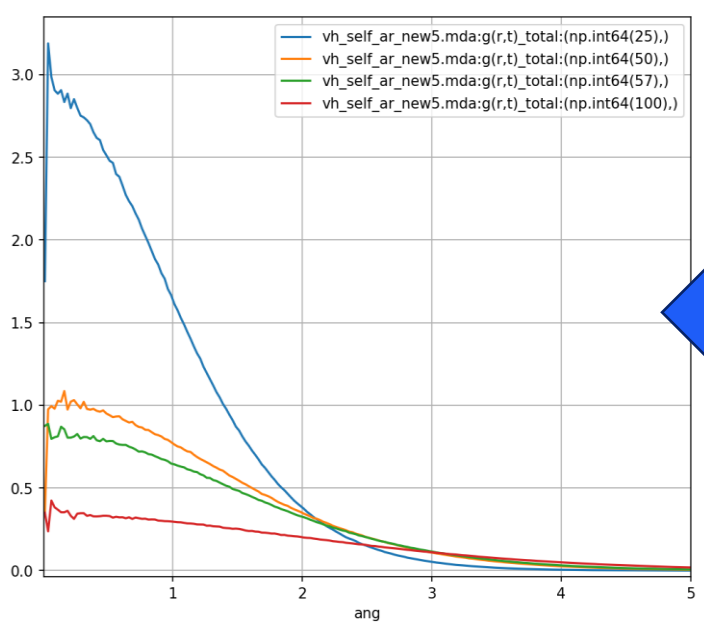
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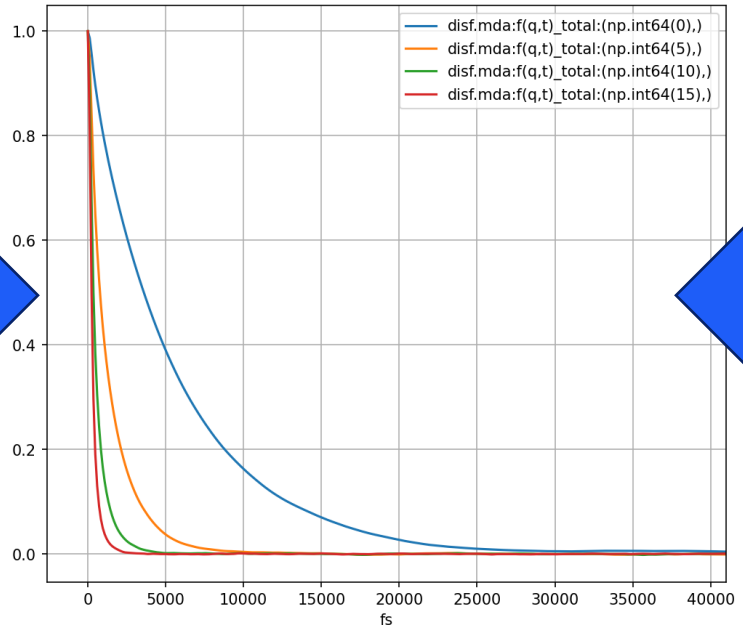
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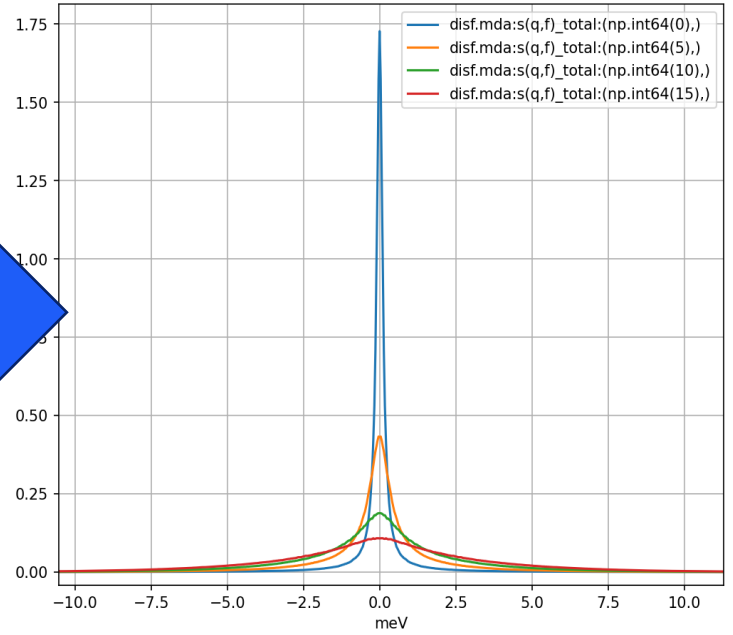
Tutorial 3: Quasielastic Neutron Scattering (QENS)

- The van Hove function (and therefore the PDF) are closely related to the dynamics structure factor via Fourier transforms. Here we show the self-part and its Fourier transforms.

$$G_{\mathrm{s}}(\mathbf{r},t) \qquad F_{\mathrm{s}}(\mathbf{Q},t) \qquad S_{\mathrm{i}}(\mathbf{Q},\omega)$$







Tutorial 3: Quasielastic Neutron Scattering (QENS)

- From the previous slides, we can see that the self-part of the van Hove function (and therefore the self intermediate scattering function and the incoherent scattering) is closely related to the diffusion of a particle.
- We can rewrite the self intermediate scattering function as the exponential of a power-series in terms of the cumulants of d(t).

$$G_{\mathrm{s}}(\mathbf{r},t) = \frac{1}{N}\sum_j \langle \delta[\mathbf{r} - \mathbf{r}_j(t_0+t) + \mathbf{r}_j(t_0)]\rangle_{t_0}$$

$$F_{\mathrm{s}}(\mathbf{Q},t) = \frac{1}{N}\sum_j \langle \exp\{i\mathbf{Q}\cdot[\mathbf{r}_j(t_0+t) - \mathbf{r}_j(t_0)]\}\rangle$$

$$= \frac{1}{N}\sum_j \langle \exp\{iq\, d_j(t)\}\rangle$$

$$d_j(t) = \hat{\mathbf{Q}}\cdot[\mathbf{r}_j(t_0+t) - \mathbf{r}_j(t_0)]$$

$$F_{\mathrm{s}}(\mathbf{Q},t) = \frac{1}{N}\sum_j \exp[-q^2\rho_{j1}(t) + q^4\rho_{j2}(t) + \cdots]$$

$$\rho_{j1}(t) = \frac{1}{2!}\langle d_j^2(t)\rangle$$

$$\rho_{j2}(t) = \frac{1}{4!}\left[\langle d_j^4(t)\rangle - 3\langle d_j^2(t)\rangle^2\right]$$

Self Intermediate Scattering Function

- Taking the leading term, the self intermediate scattering function is simply a Gaussian, this is known as the Gaussian approximation.
- When the system is isotropic, <d^2(t)> is related the MSD of the particle.
- The MSD of a particle is related to the to the diffusion constant via MSD = 6D|t|. Here I drop the summation as I assume a monoatomic system.

$$F_s(\mathbf{Q},t) = \frac{1}{N}\sum_j \exp\left[-\frac{q^2}{2}\langle d_j^2(t)\rangle\right]$$

$$F_s(q,t) = \frac{1}{N}\sum_j \exp\left[-\frac{q^2}{6}\langle \mathrm{MSD}_j(t)\rangle\right]$$

$$F_s(q,t) = \exp\left[-Dq^2|t|\right]$$

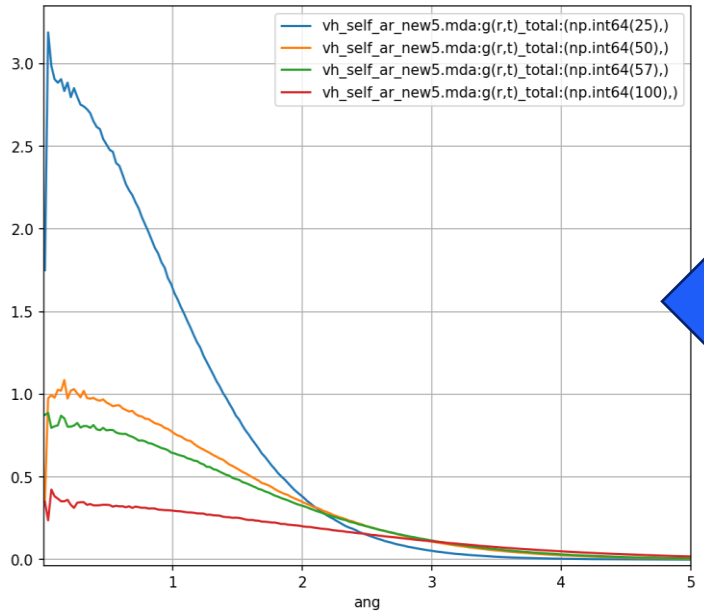
Van Hove Function (relation to diffusion)

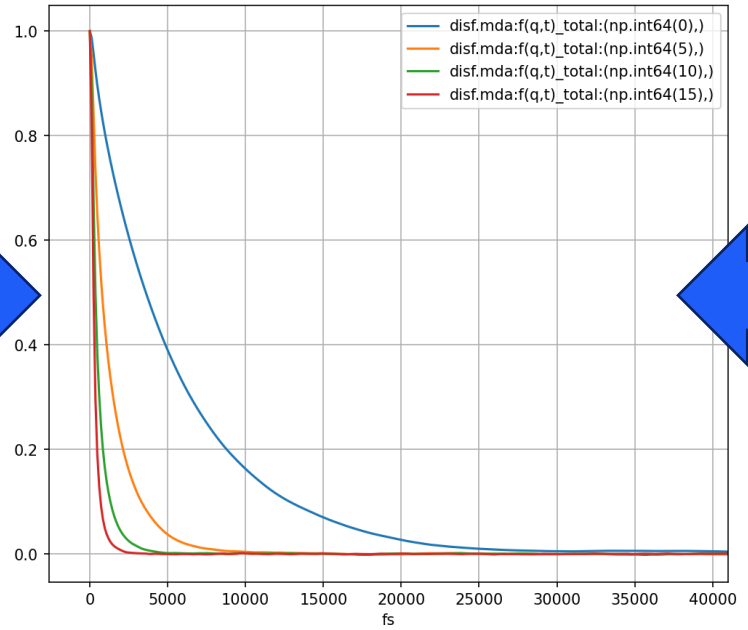
- For an isotropic system which follows the Gaussian approximation, we can see that van Hove function simply depends on the diffusion constant of the system.

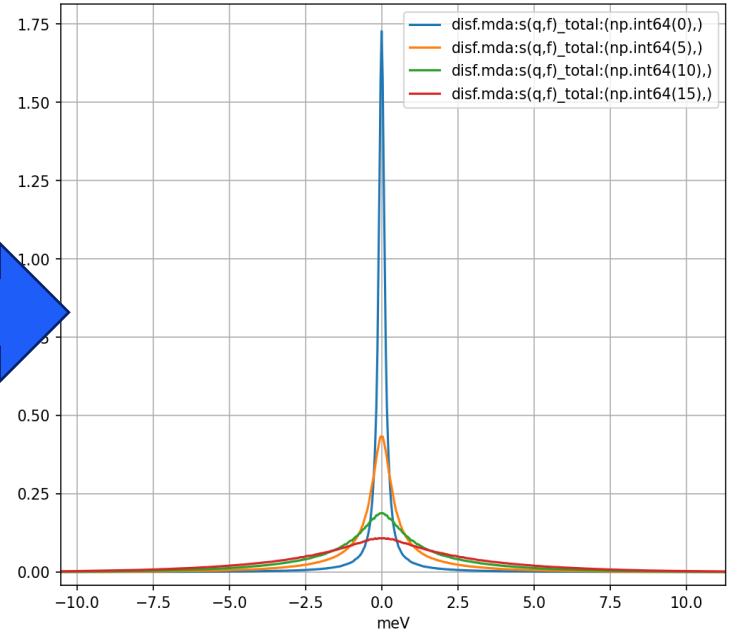
$$G_{\mathrm{s}}(r,t) = \left(\frac{1}{4\pi D|t|}\right)^{3/2} \exp\left(-\frac{r^2}{4D|t|}\right)$$

$$F_{\mathrm{s}}(q,t) = \exp\left[-Dq^2|t|\right]$$

$$S_{\mathrm{i}}(q,\omega) = \frac{1}{\pi}\frac{\hbar D q^2}{(\hbar\omega)^2 + (\hbar D q^2)^2}$$







Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Elastically scattered neutron have the same energy before and after scattering. This leads to a sharp peak at ω=0.
- Quasielastic scattering is a special case of inelastic scattering and leads to a broad peak around ω=0. QENS that can be used to determine diffusion constant of liquids and solids.
- We'll be looking at a paper by Sköld et al.

PHYSICAL REVIEW A VOLUME 6, NUMBER 3 SEPTEMBER 1972

Coherent- and Incoherent-Scattering Laws of Liquid Argon†

K. Sköld,* J. M. Rowe, and G. Ostrowski
Solid State Science Division, Argonne National Laboratory, Argonne, Illinois 60439
and
P. D. Randolph‡
Nuclear Technology Division, Idaho Nuclear Corporation, Idaho Falls, Idaho
(Received 17 January 1972)

Phys. Rev. A **6**, 1107 – **1972**
DOI: https://doi.org/10.1103/PhysRevA.6.1107

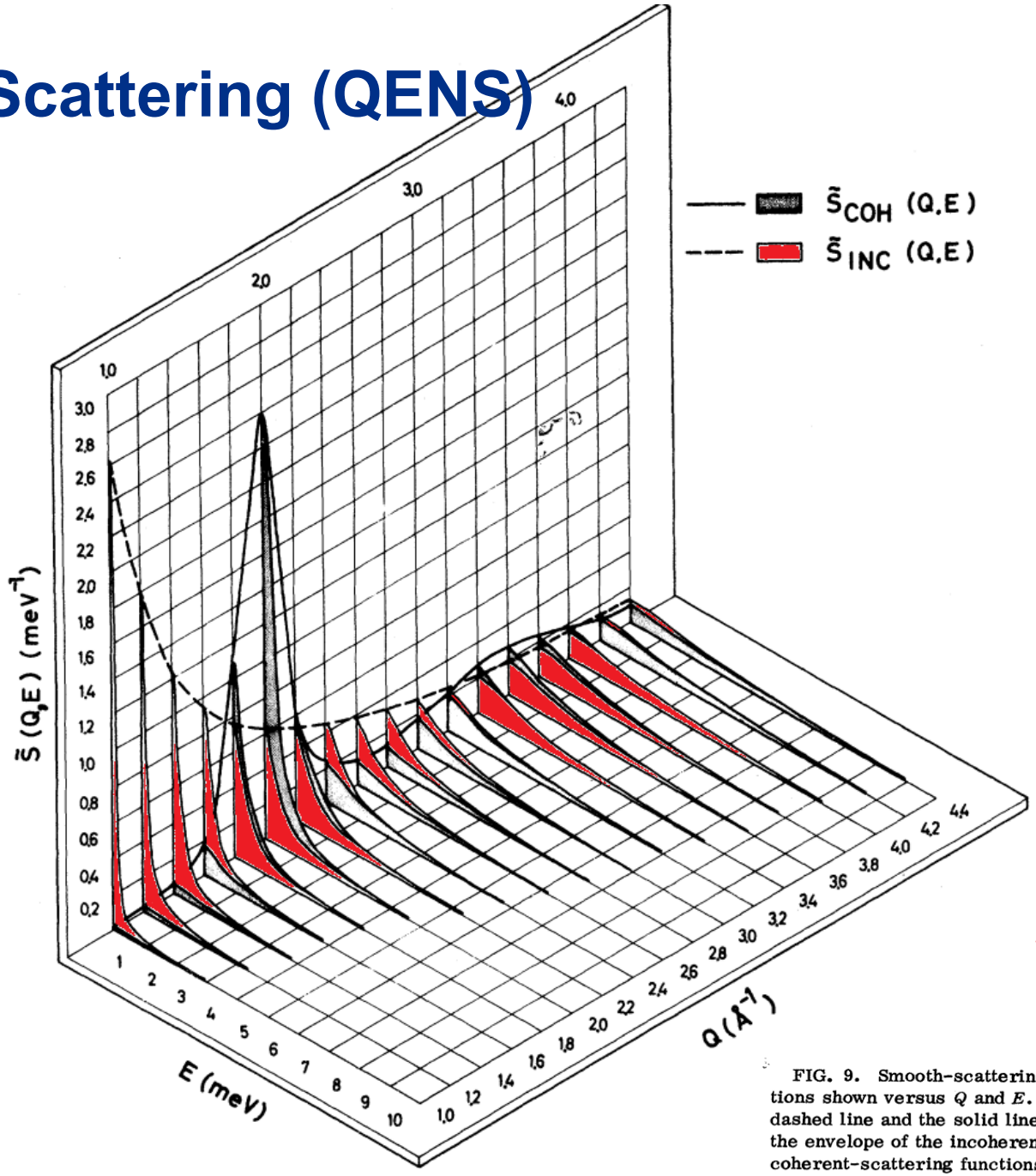


FIG. 9. Smooth-scattering functions shown versus Q and E. The dashed line and the solid line show the envelope of the incoherent and coherent-scattering functions, respectively.

Tutorial 3: Quasielastic Neutron Scattering (QENS)

- In this paper, neutrons were obtained from a reactor source (Materials Testing Reactor - Idaho National Laboratory – decommissioned 2011).

Experimental setup:

- Neutron instrument with a velocity selector (chopper-based)
- Time of flight inelastic instrument using thermal neutrons

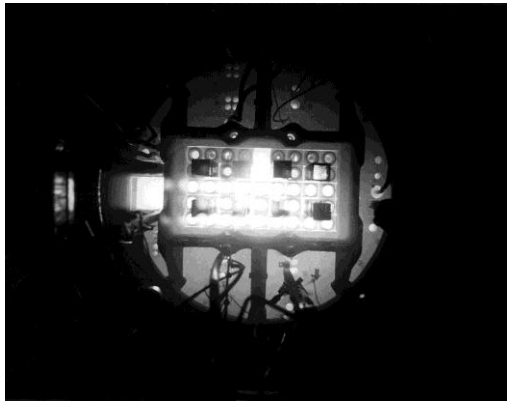


Image: View down through 20 feet of water into the tank of Materials Testing Reactor 1955

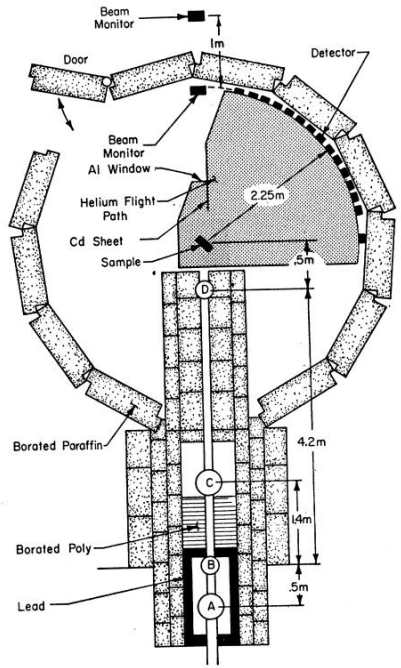


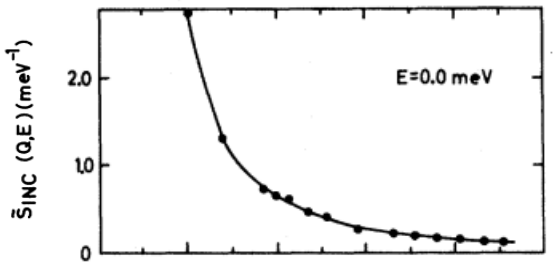
FIG. 2. Plan view of the velocity selector at the HT-1 through-hole showing the chopper arrangements, pertinent distances, and the helium-filled flight path.

Tutorial 3: Quasielastic Neutron Scattering (QENS)

- The QENS peak is a Lorentz function.
- By measuring the half-width at half-maximum of the QENS peak for a number of values of q the diffusion constant (D) of the system can be determined.

$$S_{\mathrm{i}}(q,\omega) = \frac{1}{\pi}\frac{\hbar\Gamma(q)}{(\hbar\omega)^2 + (\hbar\Gamma(q))^2}$$

$$\Gamma(q) = Dq^2$$



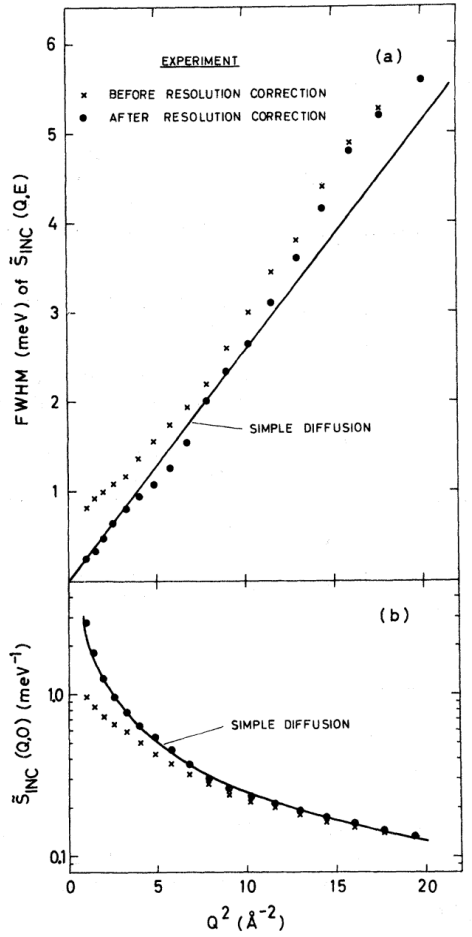


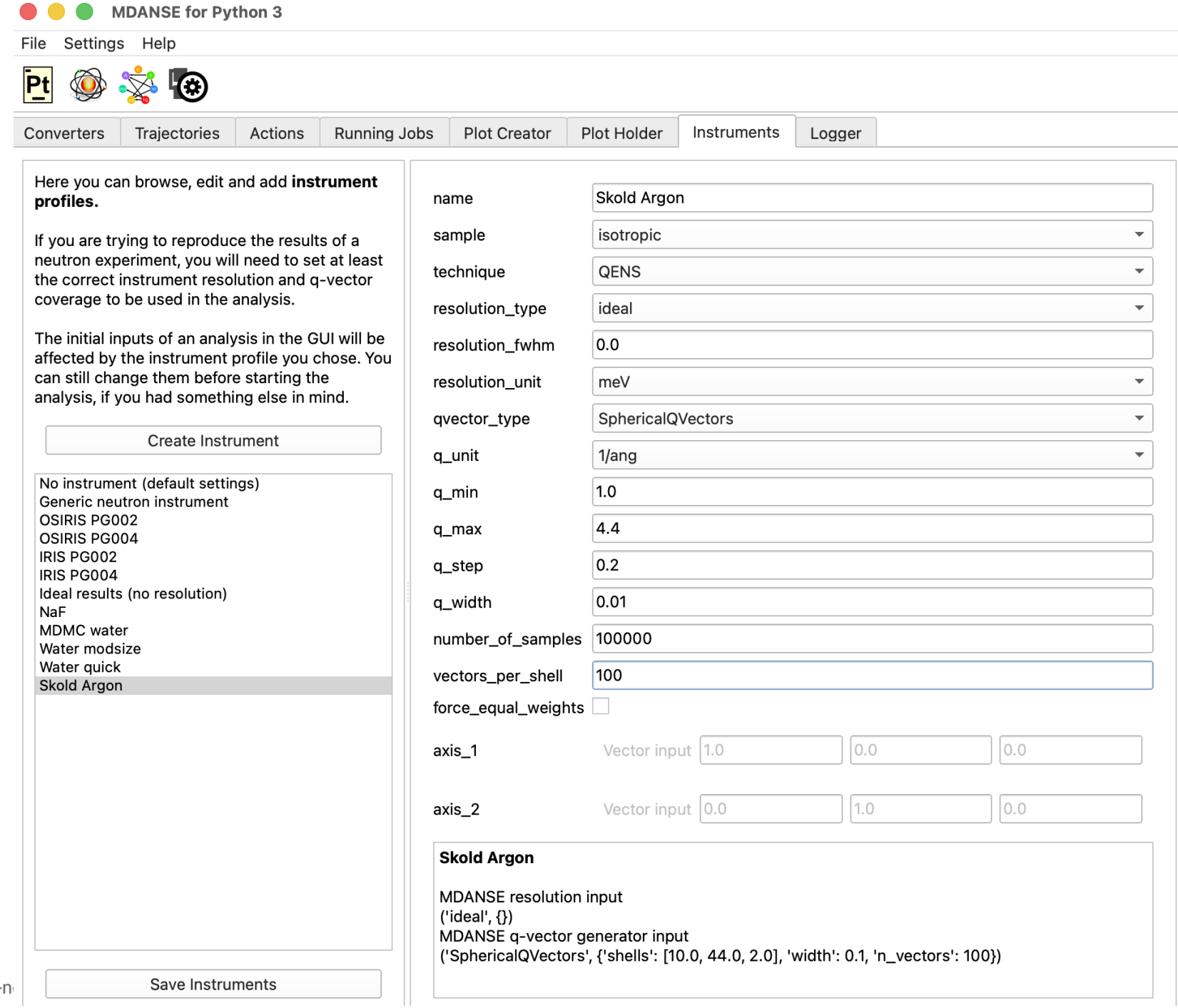
FIG. 11. (a) Full width at half-maximum of the incoherent-scattering function before and after the resolution correction. Also shown is the simple diffusion result with $D = 1.94 \times 10^{-5}$ cm^2/sec (Ref. 48). (b) Intensity at zero-energy transfer for the incoherent-scattering function before and after the resolution correction. The solid line shows the simple diffusion result.

Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Let's calculate the QENS peaks and determine the diffusion constant from it following the paper from Sköld et at.
- Load up the Argon MDT trajectory `ar36_120fs_8_f64.mdt`.
- First, we will save the scattering parameters as an instrument profile.
- Then, we will calculate the Dynamic Incoherent Structure Factor (DISF) and determine the diffusion constant by fitting the results.

Tutorial 3: Save your scattering parameters

- Go to the Instruments tab
- Click "`Create Instrument`"
- Give it a meaningful name
- Pick `resolution_type: ideal`
- Pick `qvector_type: SphericalQVectors`
- Set q values from 1.0 to 4.4 in steps of 0.2 $Å^{-1}$
- `vectors_per_shell: 100`
- `width: 0.01`
- `Save Instruments`



Tutorial 3: Quasielastic Neutron Scattering (QENS)

Go to the Actions tab.

Set the instrument to the one you just created.

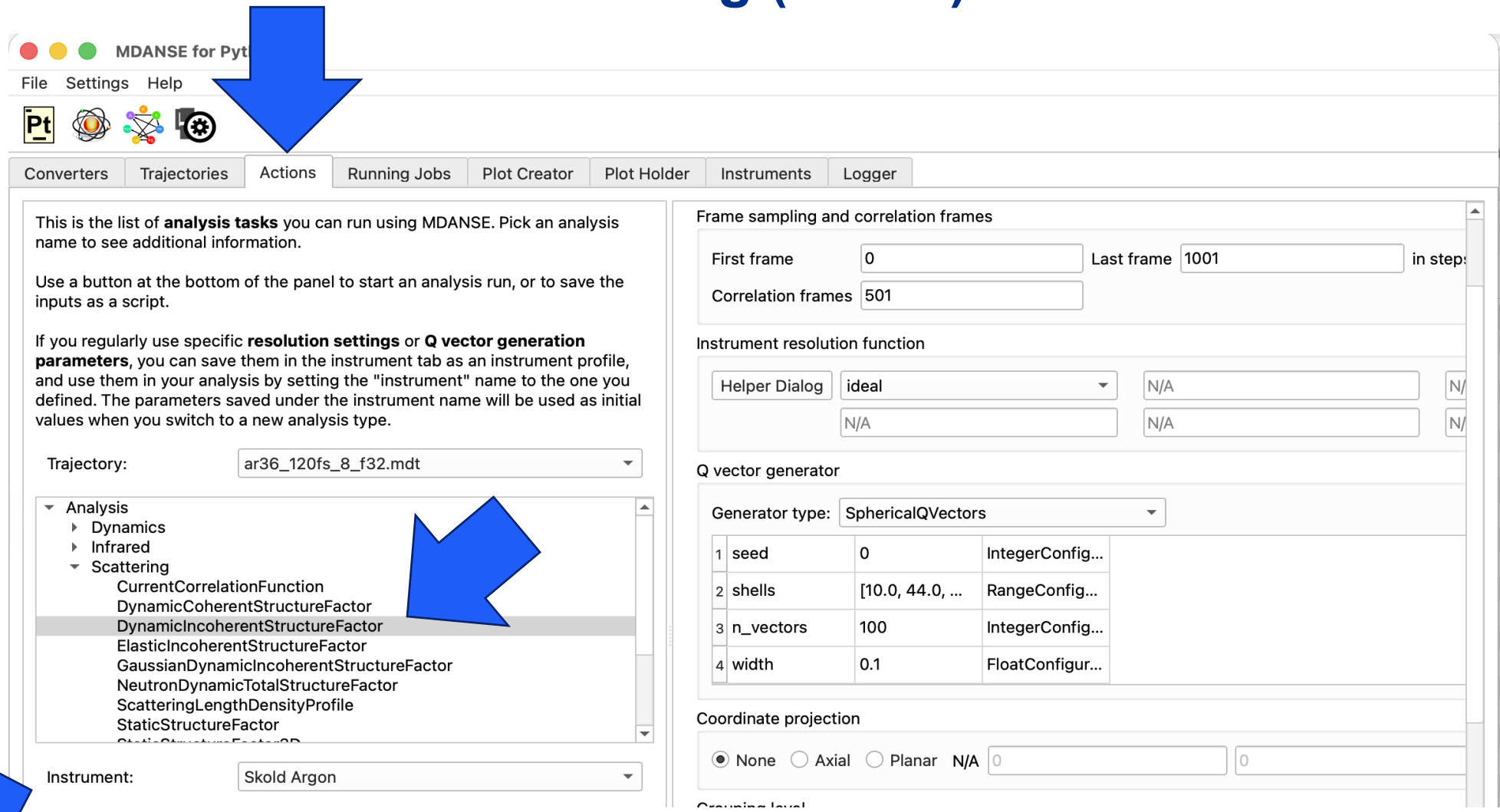
Choose:

Analysis ->

Scattering ->

DynamicIncoherent

StructureFactor



Tutorial 3: Quasielastic Neutron Scattering (QENS)

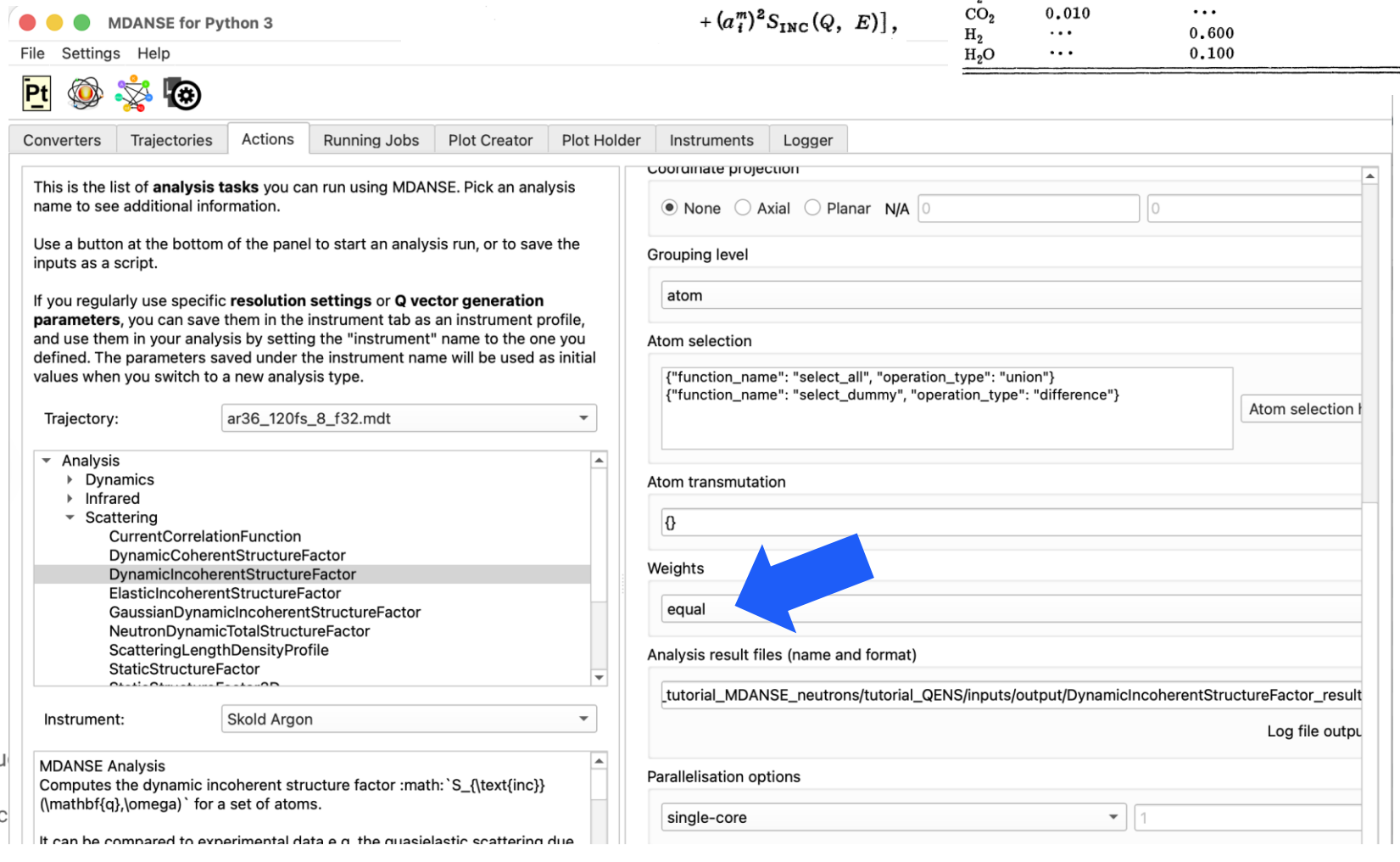
- Set the weight to equal.
- Our trajectory is pure Ar36 which does not scatter incoherently – we set equal to get an unweighted S_inc(q,f).
- Experimentally Sköld et al. obtained S_inc(q,f) by running scattering experiments on two different Argon samples and solving for S_inc(q,f).

$$S^{36}(Q,\ E) = \frac{1}{(a_c^{36})^2 + (a_i^{36})^2}\left[(a_c^{36})^2 S_{\mathrm{COH}}(Q,\ E) + (a_i^{36})^2 S_{\mathrm{INC}}(Q,\ E)\right],$$

$$S^{m}(Q,\ E) = \frac{1}{(a_c^{m})^2 + (a_i^{m})^2}\left[(a_c^{m})^2 S_{\mathrm{COH}}(Q,\ E) + (a_i^{m})^2 S_{\mathrm{INC}}(Q,\ E)\right],$$

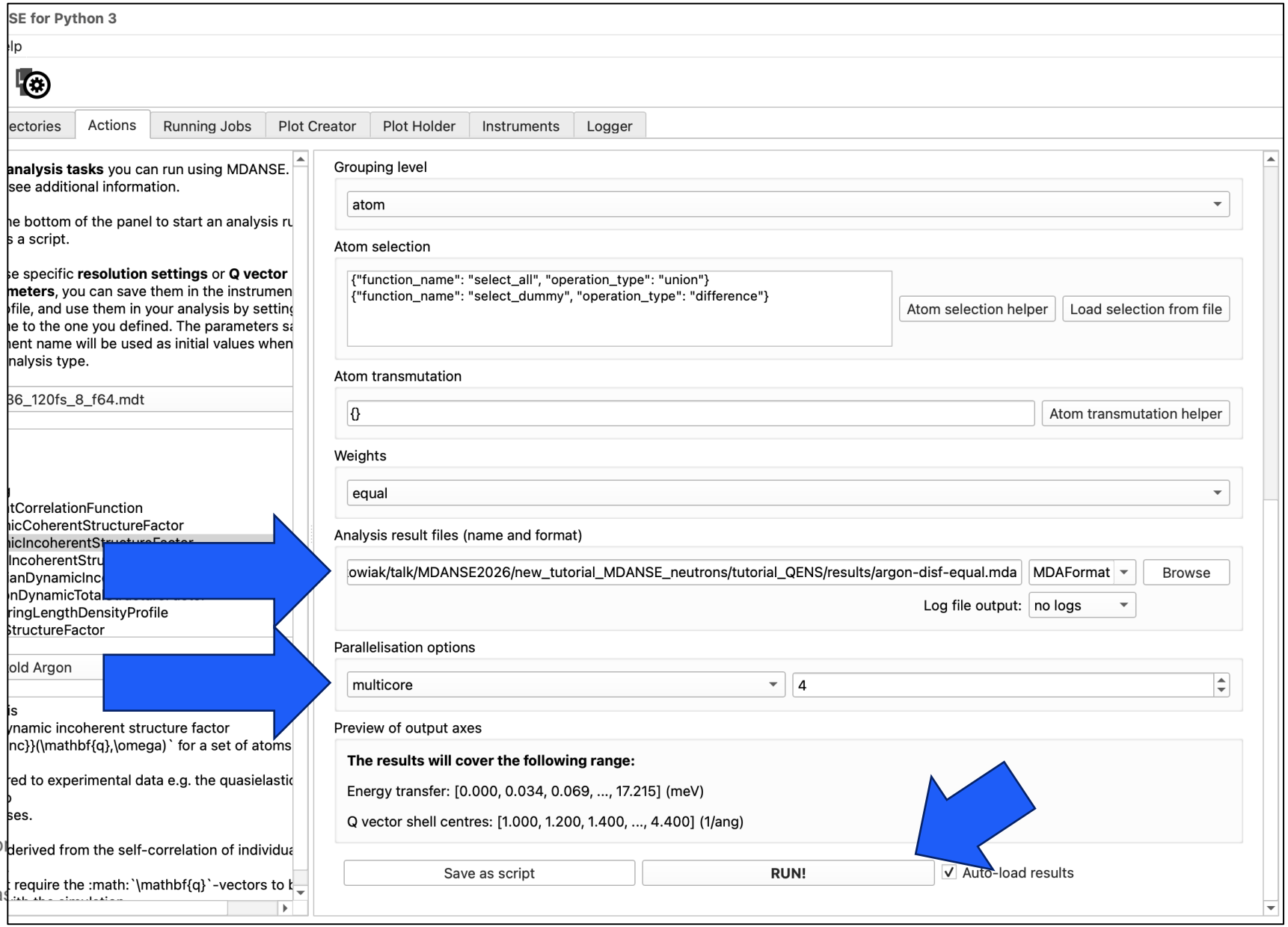
TABLE I. Composition of samples and scattering cross sections for argon isotopes.

	A^{36} sample	Mixture sample	σ_s (b)
A^{36}	98.535	7.128	77.8 ± 0.4
A^{38}	0.815	0.099	1.6 ± 1.6
A^{40}	0.080	91.773	0.419 ± 0.003
N_2	0.550	0.300	
O_2	0.010	...	
CO_2	0.010	...	
H_2	...	0.600	
H_2O	...	0.100	



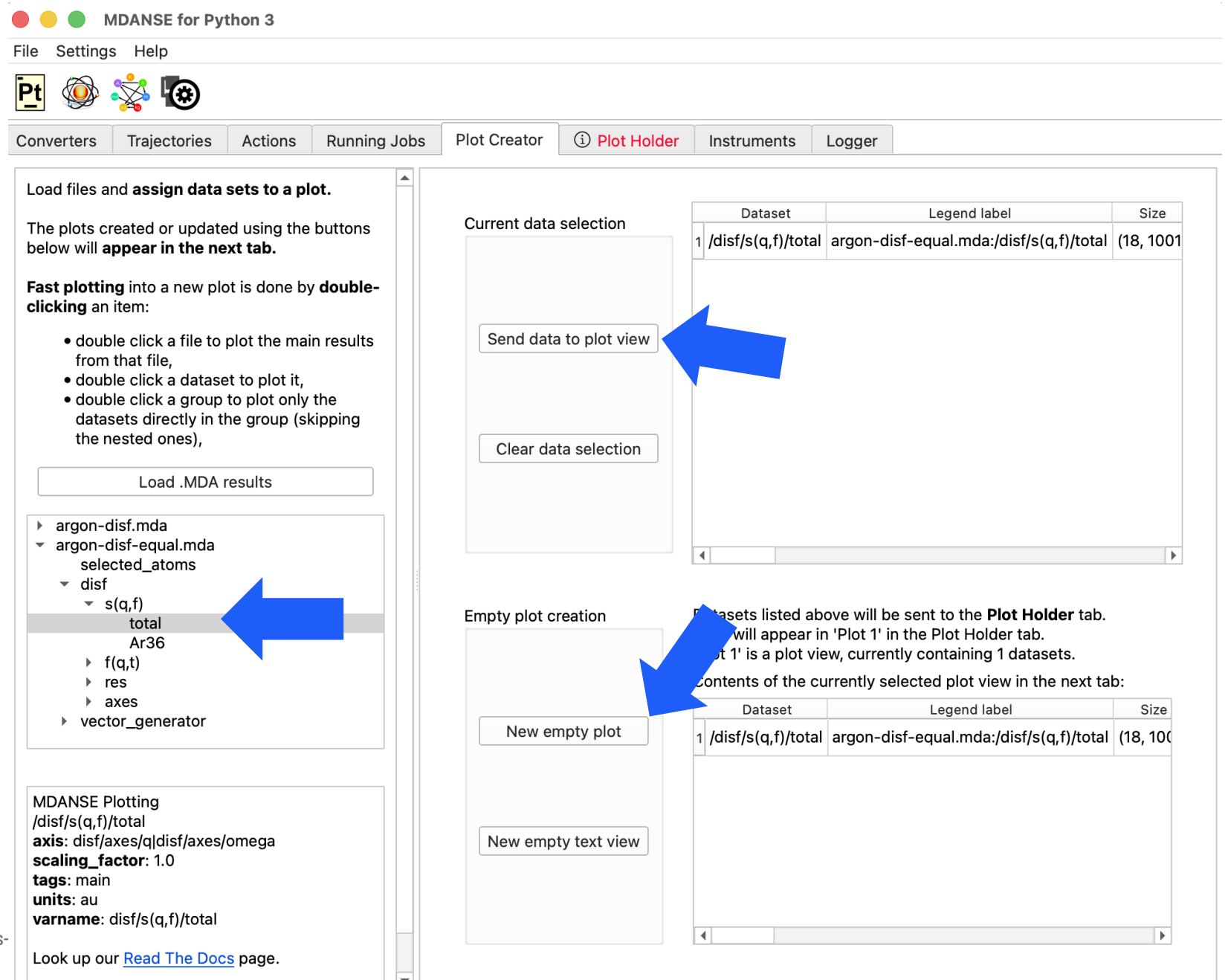
Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Set the output file, set multicore with 2 or more cores and hit “RUN!”
- Check the “Running Jobs” tab - this should take less than 3 minutes.



Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Once complete, go to the Plot Creator and create a plot of `S(q,f)_total`.



Tutorial 3: Quasielastic Neutron Scattering (QENS)

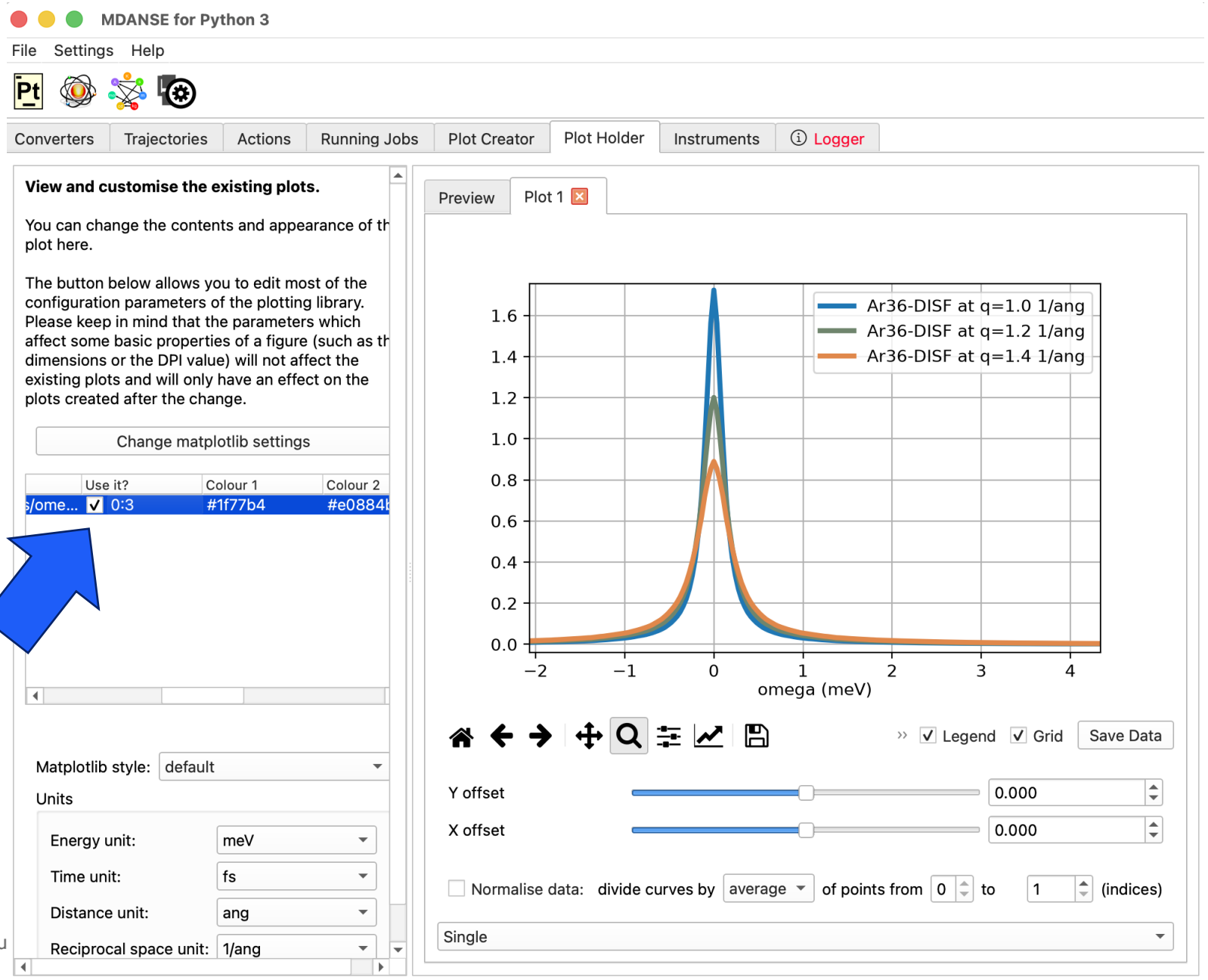
- Using your cursor determine the half width at half maximum for each plot.
- The first plot is |Q|=1.0 Å^{-1}, the next is |Q|=1.2 Å^{-1}, and etc.
- You can enter a number:

`4`

or a range:

`0:3`

in the "Use it?" column to select curves corresponding to specific `q_vector` magnitudes.



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Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Using a spreadsheet to create a plot to determine the diffusions constant.
- HINT: see FIG. 11 in the paper by Sköld et al.

$$S_{\mathrm{i}}(q,\omega) = \frac{1}{\pi}\frac{\hbar\Gamma(q)}{(\hbar\omega)^2 + (\hbar\Gamma(q))^2}$$

$$\Gamma(q) = Dq^2$$

Q (ang-1)	Q (cm-1)	Q^2 (cm-2)	HWHM (meV)	HWHM / hbar (s-1)
1				
1.2				
1.4				
1.6				
1.8				
2				
2.2				
2.4				
2.6				
2.8				
3				
3.2				
3.4				
3.6				
3.8				
4				
4.2				
4.4				

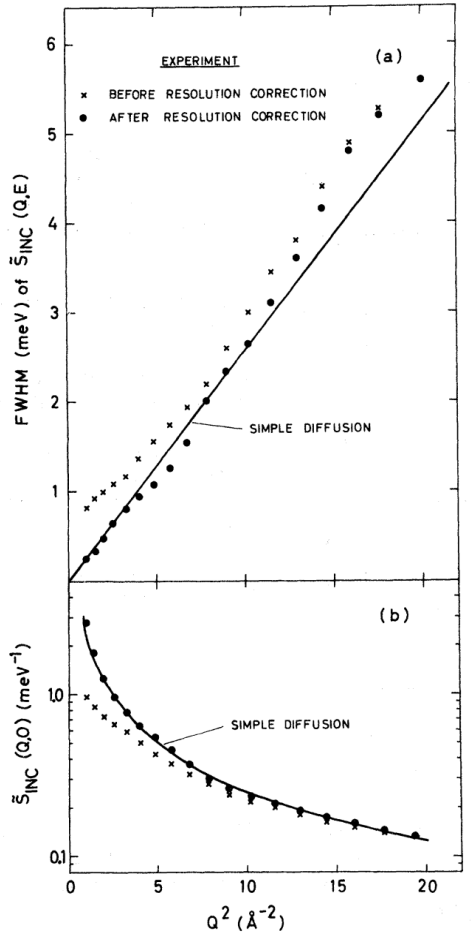
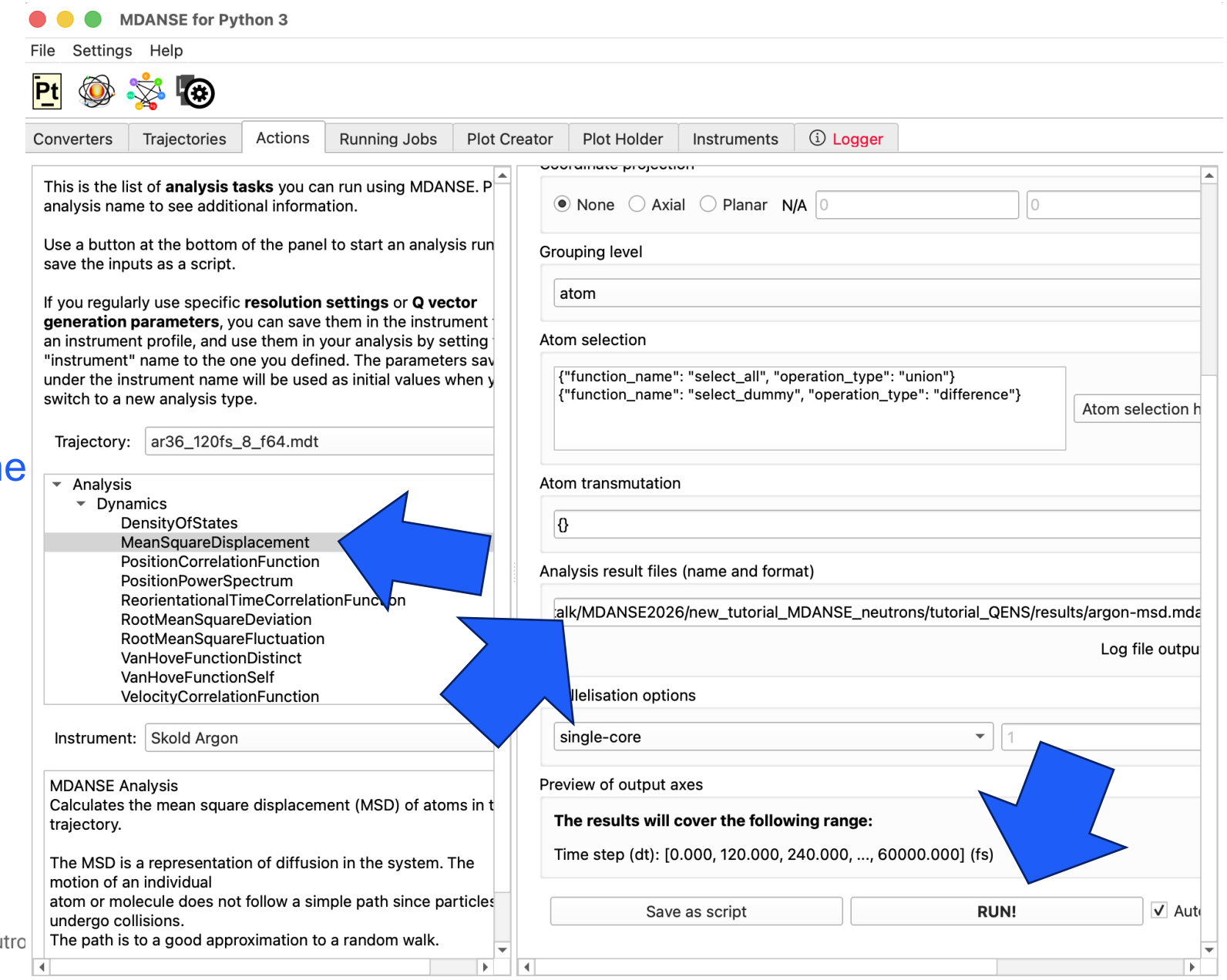


FIG. 11. (a) Full width at half-maximum of the incoherent-scattering function before and after the resolution correction. Also shown is the simple diffusion result with $D = 1.94 \times 10^{-5}$ cm^2/sec (Ref. 48). (b) Intensity at zero-energy transfer for the incoherent-scattering function before and after the resolution correction. The solid line shows the simple diffusion result.

Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Now let's calculate the diffusion constant directly from the mean square displacement and compare results.
- Using the `ar36_120fs_8_f64.mdt` run the mean square displacement calculation.
- Default settings are fine. Just set the output path and run it



Tutorial 3: Quasielastic Neutron Scattering (QENS)

- The Argon MSD is essentially linear for long times, we can calculate the diffusion constant with.

$$D = \frac{\mathrm{MSD}(t)}{6t}$$

- Calculate the diffusion constant using the last time step t=60000 fs.
- Did your results from QENS and MSD agree with the experimental?

